03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 8

(JCD 5.3 - 5.5)

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Previous lecture

- Energy transported by radiation, convection or conduction
- Assume Local Thermodynamic Equilibrium in stellar interior
- Equation of radiative transport:

$$\boxed{\frac{dT}{dr} = -\frac{3\,\kappa\rho\,L(r)}{16\pi\,a\,c\,r^2\,T^3}}$$

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Outline

Opacity

Stellar atmospheres

The energy equation

Summary

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Photon mean free path

- Remember we defined the photon mean free path as

$$\lambda = \frac{1}{\kappa_\nu \, \rho}$$

- This can also be defined in terms of cross section σ_R and the atom number density n:

$$\lambda = \frac{1}{n \, \sigma_R}$$

- Verify this relation - see the cross section as, roughly, the area around an atom where a passing photon will interact with the atom

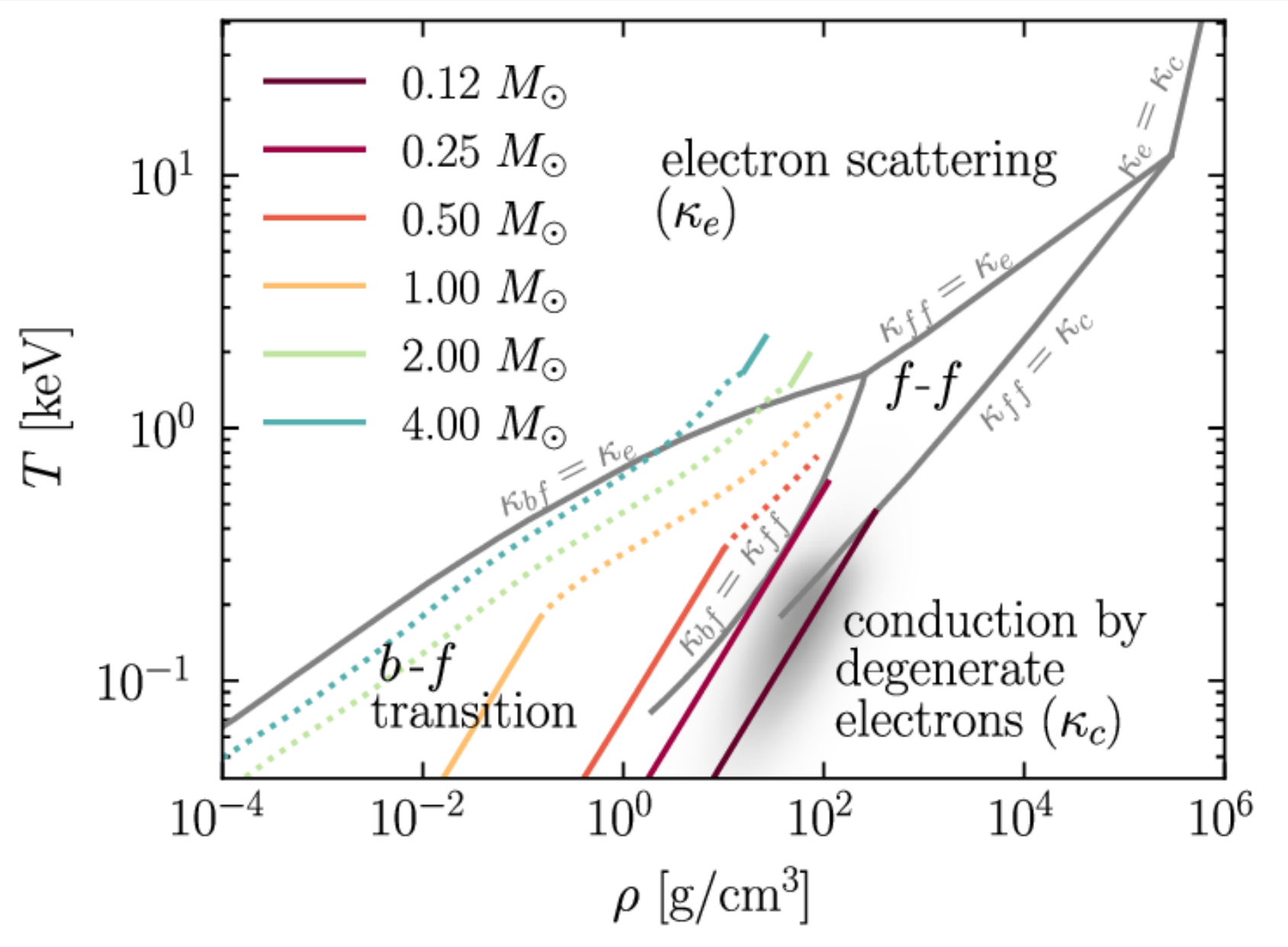
Opacity

- The opacity can thus be defined as

$$\kappa_\nu = \sigma_R \frac{n}{\rho}$$

- Interpretation: Opacity is the total cross section per unit mass
- Opacity characterises the degree to which radiation is slowed down by the presence of matter when traveling through a star
- Sources of opacity come from several types of photon-matter interactions:
  - Electron scattering: Elastic scattering of a photon by a free electron (Thomson scattering)
  - Free-free: Absorption of a photon by a free electron in the vicinity of an ion (inverse process is known as *Bremsstrahlung*)
  - Bound-free = photoionisation: photon frees a previously bound electron
  - Bound-bound: Absorption of photon excites a bound electron in an atom to a higher energy state

Main opacity sources



(Lütgert et al. 2022)

Kramers' opacity law

- Cross sections typically hard to compute
- We can make a good approximation to relate opacity with density, temperature and chemical composition
- Kramers approximation - holds for interaction between radiation and atoms:

$$\kappa_\nu \propto \rho\, T^{-7/2}$$

- The proportionality constant then depends on chemical composition:
  - Bound-free:

$$\kappa_\nu \propto Z(1+X)\rho\, T^{-7/2}$$

  - Free-free:

$$\kappa_\nu \propto (1-Z)(1+X)\rho\, T^{-7/2}$$

Electron scattering

- In the case of electron scattering, in hot less dense environments, the opacity becomes solely dependent on chemical composition:

$$\kappa_{\nu,e} = 0.2\,(1 + X)\,\mathrm{cm}^2\,\mathrm{g}^{-1}$$

- This happens in eg more massive stars

Opacity: an active field of study

- Lots of current research
- Highly computational problem - hence why approximations used

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Computing stellar atmospheric models

- Requires detailed description of radiative transfer
- Includes all states of all individual atoms
- Stellar atmospheric light is exactly what spectroscopy measures
- Not a serious problem for stellar interior structure, but stellar models do require the $P - T$ relation in the atmosphere
- Assume atmosphere roughly isothermal
- Non-LTE effects often still ignored but are being studied

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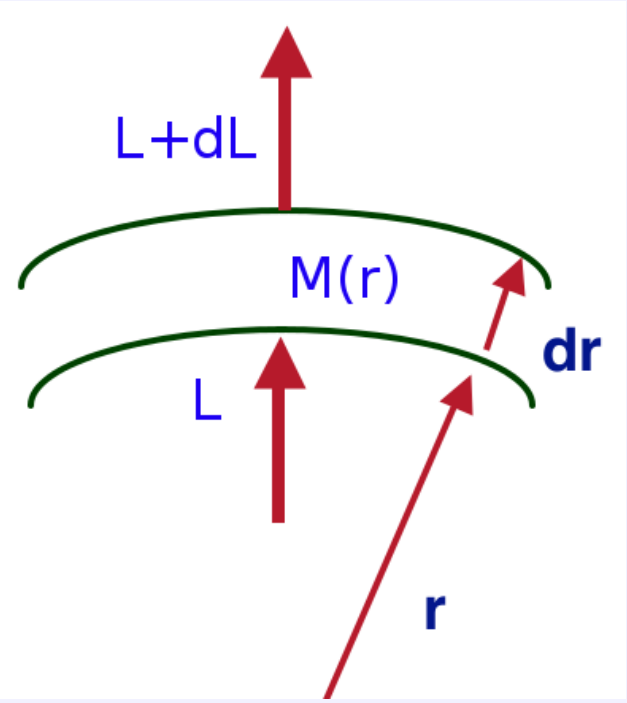
Energy generation equation - first estimate

Consider again a thin shell in the stellar interior



- Energy generated within the shell equals the energy lost through the outer surface (dL)
- Energy generated: $dM(r)\epsilon(r)$ where $\epsilon(r)$ (units of W/kg) is the energy generation rate (see later)
- Remember mass equation: $dM(r) = 4\pi\, r^2 \rho(r)\, dr$
- Energy lost = energy generated:

$$dL(r) = 4\pi\, r^2 \rho(r) \epsilon(r)\, dr$$

Energy generation equation

- Previous estimate not complete as it does not account for e.g. energy release during gravitational contraction
- Start from the first law of thermodynamics again:

$$dQ = dU + PdV$$

  with dQ the heat added or subtracted to the system
- We can show (verify!) that

$$\frac{dL}{dr} = 4\pi\, r^2 \left[\rho\epsilon - \rho\frac{d}{dt}\left(\frac{u}{\rho}\right) + \frac{P}{\rho}\frac{d\rho}{dt}\right]$$

- During nuclear burning, mostly the first term contributes.
- During gravitational contraction, only the last two terms contribute.

Summary

- Opacity can be defined as

$$\kappa_\nu = \sigma_R \frac{n}{\rho}$$

- Sources of opacity: electron scattering, free-free absorption, bound-free absorption, bound-bound absorption
- Energy equation:

$$\frac{dL}{dr} = 4\pi\, r^2 \left[\rho\epsilon - \rho \frac{d}{dt}\left(\frac{u}{\rho}\right) + \frac{P}{\rho}\frac{d\rho}{dt}\right]$$

NEXT LECTURE: Convection, instability condition, convection regions